An efficient minimization method for nonsmooth and nonconvex unconstrained optimization problems

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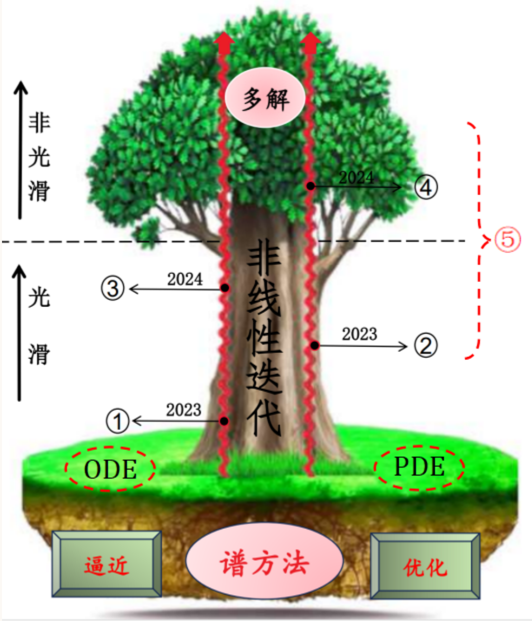


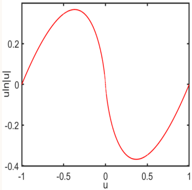
④ ⇒ 目标：高效计算非光滑的微分方程、多解、优化问题等。

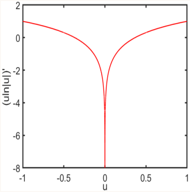
问题一：

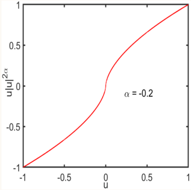
$$Q1: \quad \begin{cases} u'' + \frac{s-1}{r}u' + u\ln|u| = 0, & r \in (0,\infty), \\ u'(0) = 0, \quad (u(r), u'(r)) \to (0,0) \text{ as } r \to \infty, \end{cases} \tag{0-1}$$

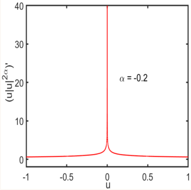
$$Q2: \quad \begin{cases} \triangle u + u\ln|u| = 0, & \text{in } \Omega = (-5,5)\times(-5,5), \\ u = 0 & \text{on } \partial\Omega. \end{cases} \tag{0-2}$$

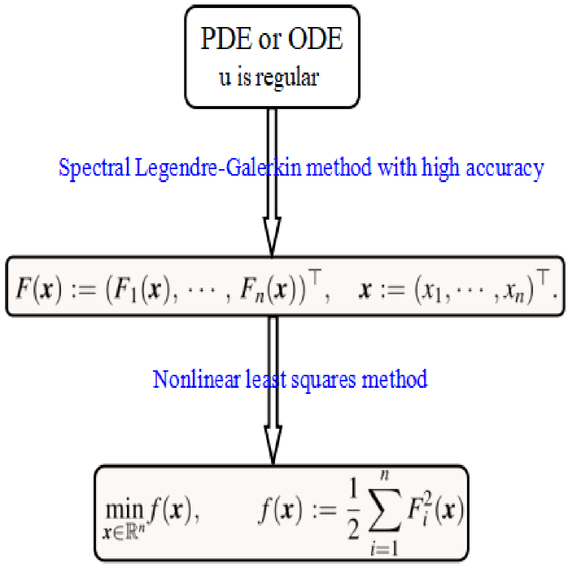
$$Q3: \quad \begin{cases} -\triangle u + \lambda|u|^{2\alpha}u = 0, & \text{in } \Omega = (-8,8)\times(-8,8), \\ u = 0 & \text{on } \partial\Omega, \end{cases} \tag{0-3}$$











注：$f(\boldsymbol{x})$ is not differentiable with respect to $\boldsymbol{x}$.

问题二：

源于回归、图像分解、视觉编码、压缩感知等优化问题的非光滑目标函数有：

$$f(\boldsymbol{x}) = \max_{1\le i\le n} \left|\sum_{j=1}^{n} \frac{x_j}{i+j-1}\right|, \tag{0-4}$$

$$f(\boldsymbol{x}) = \frac{1}{m}\|A\boldsymbol{x} - b\|_1 + \lambda\|\boldsymbol{x}\|_0, \tag{0-5}$$

$$f(\boldsymbol{x}) = \sum_{i=1}^{n-1} \max\{-x_i - x_{i+1}, -x_i x_{i+1} + (x_i^2 + x_{i+1}^2 - 1)\}, \tag{0-6}$$

$$f(\boldsymbol{x}) = \sum_{i=1}^{n-1} \max\{x_i^4 + x_{i+1}^2,\ (2-x_i)^2 + (2-x_{i+1})^2,\ 2e^{-x_i+x_{i+1}}\}. \tag{0-7}$$

注: $m, \lambda \in \mathbb{R}$，$\|\boldsymbol{x}\|_0 = \sum_{i=1, x_i\neq 0}^{n} |x_i|^0$，$b \in \mathbb{R}^n$，$A \in \mathbb{R}^{n\times n}$.

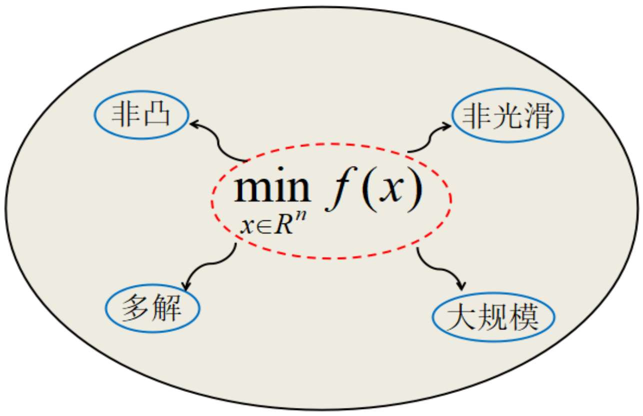


Figure 0-1 核心问题。

应用: 极小化机器学习中损失函数 := 方程 + 边界条件.

A novel method for unconstrained optimization problems

1. An efficient method for the objective function with the nonsmoothness
   - An efficient quadratic model
   - An efficient minimization method
2. Convergence analysis for the efficient minimization method
3. An auxiliary function approach for escaping local minima and computing multiple solutions.
   - An auxiliary function approach for escaping local minima
   - An auxiliary function approach for computing multiple solutions
4. Numerical experiments and solver
5. Summary and future work

An efficient quadratic model

In the trust region $\mathcal{B}_\rho(\boldsymbol{x}^{(k)}) := \left\{\boldsymbol{x} \in \mathbb{R}^n : \|\boldsymbol{x} - \boldsymbol{x}^{(k)}\|_2 \le \rho\right\}$,

$$Q_k(\boldsymbol{x}) = \tfrac{1}{2}(\boldsymbol{x} - \boldsymbol{x}^{(k)})^\top \boldsymbol{G}_k(\boldsymbol{x} - \boldsymbol{x}^{(k)}) + \boldsymbol{g}_k^\top(\boldsymbol{x} - \boldsymbol{x}^{(k)}) + c_k, \quad \forall \boldsymbol{x} \in \mathcal{B}_\rho(\boldsymbol{x}^{(k)}), \tag{1-8}$$

where $\boldsymbol{G}_k \ (= \boldsymbol{G}_k^\top) \in \mathbb{R}^{n\times n}, \boldsymbol{g}_k \in \mathbb{R}^n$ and $c_k \in \mathbb{R}$ in (1-8) are undetermined. We seek $Q_k(\boldsymbol{x})$ to be the optimiser of the following problem:

$$\begin{cases} \min\limits_{Q \in \mathbb{Q}_k} \|Q - Q_{k-1}\|_{H^2}^2 \\ \text{subject to } Q(\boldsymbol{x}_i) = f(\boldsymbol{x}_i), \quad \boldsymbol{x}_i \in \mathcal{B}_\rho(\boldsymbol{x}^{(k)}), \quad i = 1, \cdots, m. \end{cases} \tag{1-9}$$

theorem 1.1

The problem (1-9) is uniquely solvable. Moreover, within the trust region $\mathcal{B}_\rho(\boldsymbol{x}^{(k)})$, there holds

$$\|Q_k - f\|_{H^2} \le \|Q_{k-1} - f\|_{H^2}. \tag{1-10}$$

remark 1.1

1. *From* (1-10), *it can be derived that the optimal solution* Q_k *of the proposed model* (1-9) *is closer to the objective function f as the iteration proceeds (*$k \to +\infty$*). This property is instrumental in achieving solutions of higher numerical accuracy.*
2. *Compared with Powell's model (i.e.,* $\|\nabla^2(Q - Q_{k-1})\|_F^2$*), our model* (1-9) *proposed in this work exhibits several distinct advantages, which will be validated in the numerical experiments.*

Question:

- Determine the total number of points?
- Choose the set of interpolation points $\{\boldsymbol{x}_i\}_{i=1}^{m}$ given in (1-9)?

Answer:

- $2n+1$ interpolation points.
- **Algorithm 1**.

Algorithm 1. An algorithm for choosing Λ-poised interpolation points

Step 1. Choose a constant $\bar{\Lambda}$, give an interpolation set $\boldsymbol{X} = \{\boldsymbol{x}_1, \boldsymbol{x}_2, \cdots, \boldsymbol{x}_m\}$ and compute the Lagrange polynomials $\ell_i(x)$, $i = 1, 2, \cdots, m$. The iteration number is denoted as k, and $k = 1, 2, \cdots$;

Step 2. Compute $\Lambda_{k-1} = \max_{0 \leq i \leq m} \max_{x \in \Omega} |\ell_i(x)|$;

Step 3. If $\Lambda_{k-1} > \bar{\Lambda}$, then let $i_k \in \{1, \ldots, m\}$ be an index for which $\max_{x \in \Omega} |\ell_{i_k}(x)| > \bar{\Lambda}$, and let $x^*_{i_k} \in \Omega$ be a point that maximizes $|\ell_{i_k}(x)|$ in Ω, and update the interpolation set $\boldsymbol{X}$ by performing the point exchange, i.e.,

$$\boldsymbol{X} \leftarrow \boldsymbol{X} \cup \left\{x^*_{i_k}\right\} \setminus \left\{x_{i_k}\right\}.$$

Otherwise (i.e., $\Lambda_{k-1} \leq \bar{\Lambda}$), $\boldsymbol{X}$ is Λ-poised and stop;

Step 4. Update all Lagrange polynomial coefficients, and goto **Step 2**.

theorem 1.2

Within the trust region Ω, it is assumed that $f(\boldsymbol{x})$ has a bounded third-order derivative, with ν_d serving as an upper bound on its third-order derivative, and let Δ denote the diameter of Ω. Then, the following estimate holds:

$$|f(\boldsymbol{x}) - Q_k(\boldsymbol{x})| \leq \frac{1}{6}(2n+1)\nu_d \Gamma \Delta^3, \tag{1-11}$$

where

$$\Lambda_i = \max_{1 \leq j \leq m} \max_{\boldsymbol{x} \in \Omega} |\ell_j^{(i)}(\boldsymbol{x})| \tag{1-12}$$

$$\Gamma = \max\left\{1, (1 + \frac{\max_{1 \leq j \leq m} |\ell_j^{(i-1)}(\boldsymbol{x}_*^{i_k})|}{|\ell_{i_k}^{(i-1)}(\boldsymbol{x}_*^{i_k})|})\Lambda_{i-1}\right\}. \tag{1-13}$$

Example 1:

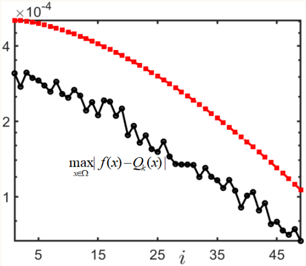
- $f(\boldsymbol{x}) = x_1^2 + 2x_2^2 + 0.3x_1x_2 + 0.1\sin(x_1 + x_2) + 0.05x_1^3 - 0.04x_2^3 + 0.02x_2^2x_1;$

Example 2:

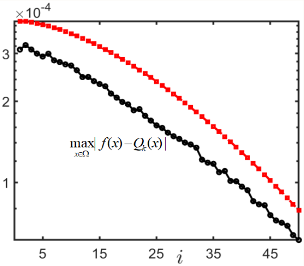
- $$f(\boldsymbol{x}) = \sum_{i=1}^{10}\{(1 + 0.1i)x_i^2 + [0.05 + 0.04(-1)^i]x_i^3\} + \sum_{i=1}^{9}\{0.3\,x_ix_{i+1} \\ +0.1\sin(x_i + x_{i+1}) + 0.02x_i^2x_{i+1}\};$$

Example 3:

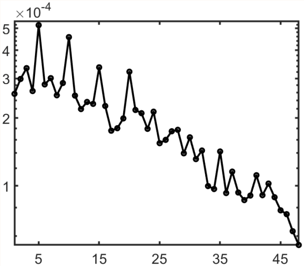
- $f(\boldsymbol{x}) = \|\boldsymbol{x}\|_1 \ln \|\boldsymbol{x}\|_1 = \sum_{i=1}^{n} |x_i| \ln \sum_{i=1}^{n} |x_i|.$



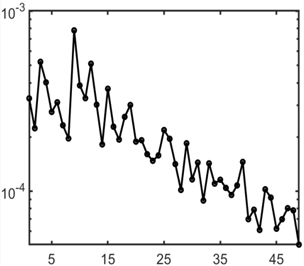
(a) *Example 1*



(b) *Example 2*



(c) *Example 3* with $n = 2$



(d) *Example 3* with $n = 10$

Within the trust region $\mathcal{B}_{\rho_k}(\boldsymbol{x}^{(k)})$, it holds that

$$\|Q_k - Q_{k-1}\|_{H^2}^2 = \rho_k^n \Psi(\boldsymbol{G}_k, \boldsymbol{g}_k, c_k), \tag{1-14}$$

where

$$\begin{aligned}\Psi(\boldsymbol{G}_k, \boldsymbol{g}_k, c_k) = &\omega_1 \|\boldsymbol{G}_k - \boldsymbol{G}_{k-1}\|_F^2 + \omega_2 \|\boldsymbol{g}_k - \boldsymbol{g}_{k-1}\|_2^2 + \omega_3 (\mathrm{Tr}(\boldsymbol{G}_k - \boldsymbol{G}_{k-1}))^2 \\ &+ \omega_4 \mathrm{Tr}(\boldsymbol{G}_k - \boldsymbol{G}_{k-1}) + (c_k - c_{k-1})^2,\end{aligned}$$

and

$$\omega_1 = \tfrac{\rho_k^4}{2(n+4)(n+2)} + \tfrac{\rho_k^2}{n+2} + 1, \quad \omega_2 = \tfrac{\rho_k^2}{n+2} + 1, \quad \omega_3 = \tfrac{\rho_k^4}{4(n+4)(n+2)}, \quad \omega_4 = \tfrac{\rho_k^2}{n+2},$$

and $\mathrm{Tr}(\cdot)$ represents the trace of a matrix. We can obtain the following constrained optimization problem to determine $\boldsymbol{G}_k$, $\boldsymbol{g}_k$ and c_k:

$$\begin{cases} \min \Psi(\boldsymbol{G}_k, \boldsymbol{g}_k, c_k) \\ \text{subject to } Q_k(\boldsymbol{x}_i) = f(\boldsymbol{x}_i), \quad 1 \le i \le m. \end{cases} \tag{1-15}$$

The corresponding Lagrange multiplier function $\mathcal{L}(\boldsymbol{G}_k, \boldsymbol{g}_k, c_k)$ is formed as follow:

$$\mathcal{L}(\boldsymbol{G}_k, \boldsymbol{g}_k, c_k) = \Psi(\boldsymbol{G}_k, \boldsymbol{g}_k, c_k) - \textstyle\sum_{l=1}^m \lambda_l (Q_k(\boldsymbol{x}_l) - f(\boldsymbol{x}_l)). \tag{1-16}$$

proposition 1.1

$\boldsymbol{G}_k$ presented in (1-8) becomes

$$\begin{aligned}\boldsymbol{G}_k =& \frac{1}{2\omega_1}\{-(\frac{\omega_3}{2(n\omega_3+\omega_1)}\sum_{l=0}^{m}\lambda_l\|\boldsymbol{x}_l-\boldsymbol{x}^{(k)}\|_2^2 \\ & \frac{1}{2}\sum_{l=1}^{m}\lambda_l(\boldsymbol{x}_l-\boldsymbol{x}^{(k)})(\boldsymbol{x}_l-\boldsymbol{x}^{(k)})^\top+\frac{\omega_1}{n\omega_3+\omega_1}\omega_4 c)\boldsymbol{I}\}+\boldsymbol{G}_{k-1}.\end{aligned} \tag{1-17}$$

$\boldsymbol{\lambda} := (\lambda_1, \lambda_2, \cdots, \lambda_m)^\top, c_k$ and $\boldsymbol{g}_k$ in (1-8) are obtained by solving (1-18):

$$\begin{pmatrix} \boldsymbol{A} & \mathrm{diag}\{\boldsymbol{J}\} & \mathrm{diag}\{\boldsymbol{Y}^\top\} \\ \boldsymbol{J}^\top & \frac{n\omega_4^2}{2(n\omega_3+\omega_1)}-2 & \boldsymbol{0}_n^\top \\ \boldsymbol{Y} & \boldsymbol{0}_n^\top & -2\omega_2\boldsymbol{I} \end{pmatrix}\begin{pmatrix}\boldsymbol{\lambda} \\ c_k \\ \boldsymbol{g}_k\end{pmatrix} = \begin{pmatrix}\mathcal{H} \\ (\frac{n\omega_4^2}{2(n\omega_3+\omega_1)}-2)c_{k-1} \\ -2\omega_2\boldsymbol{g}_{k-1}\end{pmatrix}, \tag{1-18}$$

where

$$\boldsymbol{Y} = (\boldsymbol{x}_1-\boldsymbol{x}^{(k)}, \boldsymbol{x}_2-\boldsymbol{x}^{(k)}, \ldots, \boldsymbol{x}_m-\boldsymbol{x}^{(k)}),$$

$$\boldsymbol{J} = (1-\tfrac{\omega_4}{4(n\omega_3+\omega_1)}\|\boldsymbol{x}_1-\boldsymbol{x}^{(k)}\|_2^2, \cdots, 1-\tfrac{\omega_4}{4(n\omega_3+\omega_1)}\|\boldsymbol{x}_m-\boldsymbol{x}^{(k)}\|_2^2)^\top,$$

$$(\boldsymbol{A})_{ij} = \tfrac{1}{8\omega_1}[(\boldsymbol{x}_i-\boldsymbol{x}^{(k)})^\top(\boldsymbol{x}_j-\boldsymbol{x}^{(k)})]^2-\tfrac{\omega_3}{8\omega_1(n\omega_3+\omega_1)}\|\boldsymbol{x}_i-\boldsymbol{x}^{(k)}\|_2^2\|\boldsymbol{x}_j-\boldsymbol{x}^{(k)}\|_2^2,$$

$$\mathcal{H}\in\mathbb{R}^m \text{ and } (\mathcal{H})_i = f(\boldsymbol{x}_i)+(1-\tfrac{\omega_4}{4(n\omega_3+\omega_1)}\|\boldsymbol{x}_1-\boldsymbol{x}^{(k)}\|_2^2)c_{k-1}+(\boldsymbol{x}_i-\boldsymbol{x}^{(k)})^\top\boldsymbol{g}_{k-1}.$$

An efficient minimization method

Algorithm 2 An efficient minimization method for computing nonconvex and nonsmooth (1.1)

Input:

$\boldsymbol{x}^{(0)} \leftarrow$ The initial guess

$\varepsilon_{\text{glob}} \leftarrow$ Global minimum tolerance

$Q_0(\boldsymbol{x}) \leftarrow$ The initial quadratic model

$\eta, \gamma_1, \gamma_2, \mu \leftarrow$ Other parameters

$\varepsilon_{\text{stop}} \leftarrow$ The termination tolerance

$\Delta_{\max} \leftarrow$ The maximum trust region radius

$\Delta_0 \in (0, \Delta_{\max}] \leftarrow$ The initial trust region radius

Output: The solution, denoted by $\boldsymbol{x}_{\text{glob}}$

Step 1. (Initialization) Choose the initial guess $\boldsymbol{x}^{(0)}, 0 < \Delta_0 \leq \Delta_{\max}, Q_0(\boldsymbol{x}) := 0$. Sampling points $\mathcal{X}_0 \subset \boldsymbol{B}_{\Delta_0}(\boldsymbol{x}^{(0)})$ are chosen randomly, and the number of sampling points m is $2n+1$. Set $\varepsilon_{\text{glob}}, \varepsilon_{\text{stop}} \ll 1$, $0 < \eta, \omega < 1$ and $0 < \gamma_2 < 1 < \gamma_1$.

For $k = 0, 1, \ldots$ **do** ▷ Nonlinear iteration

Step 2. (Determine the Λ-poised interpolation set and the quadratic model) If $\mathcal{X}_k$ is Λ-poised in $\boldsymbol{B}_{\Delta_k}(\boldsymbol{x}^{(k)})$, the interpolating points set $\mathcal{X}_k$ is accepted. Otherwise, based on the **Algorithm** 1, the interpolating points set $\mathcal{X}_k$ is adjusted until $\mathcal{X}_k$ is Λ-poised in $\boldsymbol{B}_{\Delta_k}(\boldsymbol{x}^{(k)})$. Then, with (3.13)-(3.24), the quadratic model Q_k is constructed. Moreover, when $\Delta_k > \mu \|\boldsymbol{g}_k\|$, the following process is carried out to form a fully linear model Q_k:

Set $j = 0$, $\tilde{\Delta}_k := \Delta_k$, $\boldsymbol{g}_k^{(0)} := \boldsymbol{g}_k$ and $Q_k^{(0)} := Q_k$.

while $\tilde{\Delta}_k > \mu \|\boldsymbol{g}_k^{(j)}\|$ **do**

Set $j := j+1$ and $\tilde{\Delta}_k := \omega^j \Delta_k$,

Improve $\mathcal{X}_k$ until it is Λ-poised in $\boldsymbol{B}_{\tilde{\Delta}_k}(\boldsymbol{x}^{(k)})$,

Construct an updating model $Q_k^{(j)}$ based on $\mathcal{X}_k$.

end while

Denote $Q_k \leftarrow Q_k^{(j)}$ and $\Delta_k \leftarrow \tilde{\Delta}_k$.

Step 3. **(Minimize Q_k within the trust region Δ_k)** Compute the trust-region step $\boldsymbol{s}_k$ by solving the following trust-region subproblem:

$$\begin{cases} \min Q_k(\boldsymbol{x}^{(k)} + \boldsymbol{s}) \\ \text{subject to } \|\boldsymbol{s}\|_2 \le \Delta_k. \end{cases}$$

Compute $f(\boldsymbol{x}^{(k)} + \boldsymbol{s}_k)$ and r_k defined in (3.25). If $r_k \ge \eta$, then let $\boldsymbol{x}^{(k+1)} := \boldsymbol{x}^{(k)} + \boldsymbol{s}_k$. Take $\boldsymbol{x}^{(k+1)}$ into the interpolation set, i.e.,

$$\mathcal{X}_{k+1} := \mathcal{X}_k \cup \{\boldsymbol{x}^{(k+1)}\} \backslash \{\arg\max_{\boldsymbol{x}} \|\boldsymbol{x} - \boldsymbol{x}^{(k+1)}\|_2\}.$$

Otherwise, set $Q_{k+1} := Q_k$, $\boldsymbol{x}^{(k+1)} := \boldsymbol{x}^{(k)}$, and update the trust-region radius:

$$\Delta_{k+1} := \begin{cases} \min\{\gamma_1 \Delta_k, \Delta_{\max}\} & \text{if } r_k \ge \eta, \\ \gamma_2 \Delta_k & \text{if } r_k < \eta. \end{cases}$$

If $\|\boldsymbol{g}_{k+1}\| \ge \varepsilon_{\text{stop}}$ or $\Delta_{k+1} \ge \varepsilon_{\text{stop}}$, go to **Step 2**. Otherwise, when $|f(\boldsymbol{x}^{(k+1)})| \ge \varepsilon_{\text{glob}}$, the strategy given in section 4 is used to escape the local minima $\boldsymbol{x}^{(k+1)}$, and go to **Steps 2-3** until $|f(\boldsymbol{x}^{(k+1)})| < \varepsilon_{\text{glob}}$. Denote $\boldsymbol{x}_{\text{glob}} := \boldsymbol{x}^{(k+1)}$ and stop.

End

Return $\boldsymbol{x}_{\text{glob}}$

Convergence analysis

lemma 2.1

Suppose that for any $\boldsymbol{x} \in \mathcal{B}_{\rho_k}(\boldsymbol{x}^{(k)})$,

$$|f(\boldsymbol{x}) - Q_k(\boldsymbol{x})| \leq \kappa_f \rho_k^2, \tag{2-20}$$

and the trust-region radius at $k-$th iteration satisfies

$$\rho_k \leq \min\left\{\frac{\|\boldsymbol{g}_k\|}{\|\boldsymbol{G}_k\|}, \frac{\kappa(1-\eta)\|\boldsymbol{g}_k\|}{4\kappa_f}\right\}, \tag{2-21}$$

where κ_f is a positive constant and κ is the Cauchy decrease constant. Then the $k-$th iteration is successful, i.e., $r_k \geq \eta$.

lemma 2.2

Suppose that there exists a positive constant c_1 such that $\|\boldsymbol{g}_k\| \geq c_1$. Then there exists a positive constant c_2 such that $\rho_k \geq c_2$.

remark 2.1

By the contrapositive of the ***Lemma 2.2****, we obtain the implication*

$$\rho_k \to 0 \quad \Rightarrow \quad \boldsymbol{g}_k \to 0, \tag{2-22}$$

which will be used in the subsequent convergence analysis.

theorem 2.1

Suppose that for any $\boldsymbol{x}, \boldsymbol{y} \in \mathcal{B}_{\rho_k}(\boldsymbol{x}^{(k)})$, $\nabla f(\boldsymbol{x})$ satisfies the Lipschitz condition, i.e., there exists a constant $L > 0$, the following inequality holds:

$$|\nabla f(\boldsymbol{x}) - \nabla f(\boldsymbol{y})| \leq L\|\boldsymbol{x} - \boldsymbol{y}\|, \quad \|\nabla f(\boldsymbol{x}) - \boldsymbol{g}_k\| \leq \kappa_g \rho_k \tag{2-23}$$

where κ_g is a positive constant. Then, $\nabla f(\boldsymbol{x}^{(k)}) \to 0$ as $k \to \infty$.

An auxiliary function approach

Two highly challenging problems naturally arise as follows:

- How can a point $\boldsymbol{x}^*$ satisfying $f(\boldsymbol{x}^*) = 0$ be obtained from escaping local minima?
- How can we compute distinct points $\boldsymbol{x}^*$ satisfying $f(\boldsymbol{x}^*) = 0$, i.e., multiple solutions?

Here we are the first to propose an auxiliary function approach to address these problems. Our core idea is to introduce an auxiliary function Φ and form a new objective function:

$$\hat{f}(\boldsymbol{x}) := \Phi f(\boldsymbol{x}), \tag{3-24}$$

where Φ will be determined later. **Algorithm 2** is reused to iterate $\hat{f}(\boldsymbol{x})$ to achieve our goal.

We require that the objective function $\hat{f}(\boldsymbol{x})$ given in (3-24) satisfies following conditions:

- for a point $\boldsymbol{r}^*$, it holds that

$$f(\boldsymbol{r}^*) = 0 \quad \Leftrightarrow \quad \hat{f}(\boldsymbol{r}^*) = 0, \tag{3-25}$$

  indicating that $f(\boldsymbol{x})$ and $\hat{f}(\boldsymbol{x})$ share the same optimal solution.

- $\boldsymbol{r}$ is not a local minimum point of $\hat{f}(\boldsymbol{x})$, i.e.,

$$\nabla \hat{f}(\boldsymbol{r}) \neq \boldsymbol{0}. \tag{3-26}$$

From (3-25), we can minimize $\hat{f}(\boldsymbol{x})$ instead of the original objective function $f(\boldsymbol{x})$. Consequently, we can derive

$$\Phi(\boldsymbol{r}^*) \neq 0. \tag{3-27}$$

On the other hand, based on (3-26), we can further derive

$$\nabla \Phi(\boldsymbol{r}) \neq \boldsymbol{0} \quad \Leftrightarrow \quad \nabla \hat{f}(\boldsymbol{r}) \neq \boldsymbol{0}. \tag{3-28}$$

In addition, for any sequence $\{\boldsymbol{x}^{(j)}\}$ near the local minimum point $\boldsymbol{r}$, we require that

$$\Phi \to \infty \quad \text{as} \quad \boldsymbol{x}^{(j)} \to \boldsymbol{r}. \tag{3-29}$$

We adopt the following specific form throughout this paper:

$$\Phi(\boldsymbol{x}; \boldsymbol{r}) = \frac{1}{\|\boldsymbol{x} - \boldsymbol{r}\|_2^2} + 1. \tag{3-30}$$

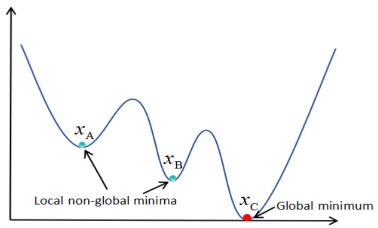


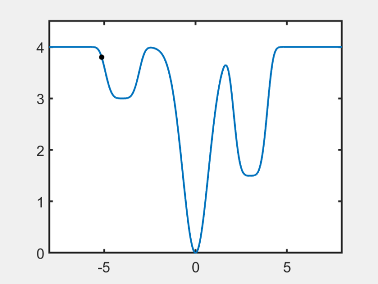
Figure 3-2 A configuration for local and global minima.

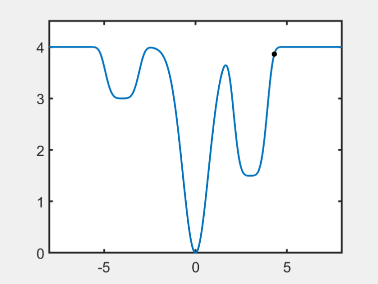
Step 1: Minimize $f(\boldsymbol{x})$ using **Algorithm 2** to obtain a local minimum $\boldsymbol{x}_{\mathrm{A}}$ (or $\boldsymbol{x}_{\mathrm{B}}$, see Fig. 3-2).

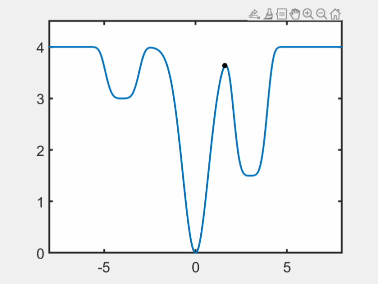
Step 2: Define a new objective function $\hat{f}(\boldsymbol{x}) := \Phi(\boldsymbol{x}; \boldsymbol{x}_{\mathrm{A}}) f(\boldsymbol{x})$.

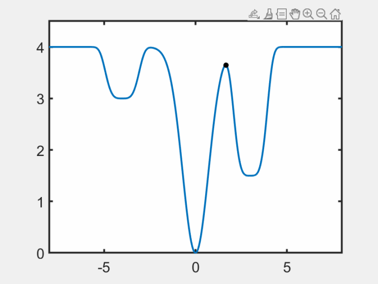
Step 3: Choose $\boldsymbol{x}_{\mathrm{A}} + \boldsymbol{\epsilon}$ as a new starting point, where $\boldsymbol{\epsilon}$ is a random perturbation vector. Reapply **Algorithm 2** to minimize $\hat{f}(\boldsymbol{x})$. If another local minimum (e.g., $\boldsymbol{x}_B$) is computed, repeat **Steps 2** and **3** until a point $\boldsymbol{x}^*$ satisfying $f(\boldsymbol{x}^*) = 0$ (e.g., $\boldsymbol{x}_C$ in Fig. 3-2) is obtained.

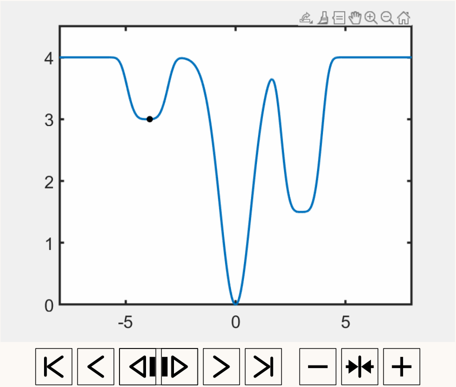
Case 1:

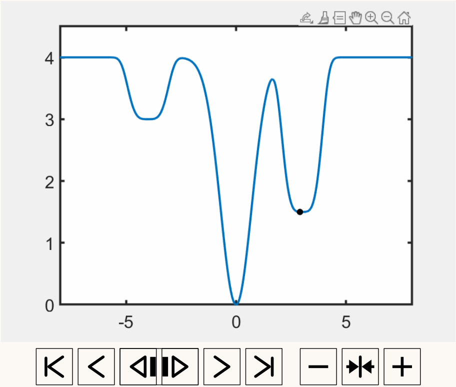




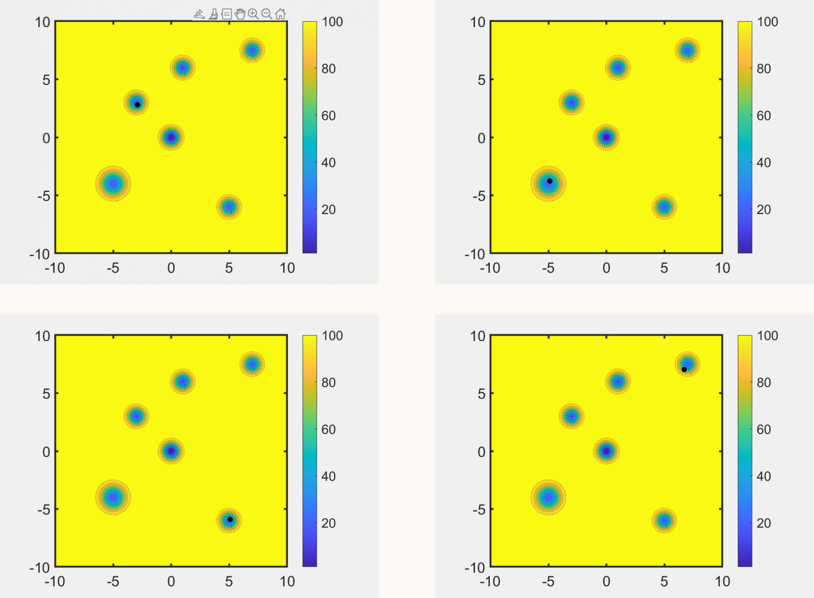





4
3
2
1
0
-5
0
5


4
3
2
1
0
-5
0
5

Case 2:



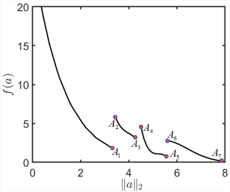
Case 3:

$$\begin{cases} u'' + \frac{s-1}{r}u' + u\ln|u| = 0, & r \in (0,\infty), \\ u'(0) = 0, \qquad (u(r), u'(r)) \to (0,0) \text{ as } r \to \infty, \end{cases} \tag{3-31}$$

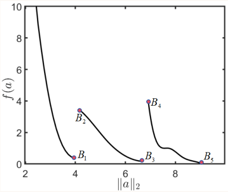
where $s > 1$.

$$\mathrm{IG}_1 = 0.1 \times ones(1, N), \qquad \mathrm{IG}_2 = 10\sin z, \qquad \mathrm{IG}_3 = e^z, \tag{3-32}$$

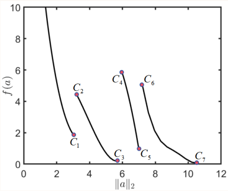
where $z = \texttt{linspace}(0, 1, N)$ and $N = 24$.



(a) IG_1



(b) IG_2



(c) IG_3

An auxiliary function approach for multiple solutions

definition 3.1

Let $\mathbb{X}$ be a n-dimensional Euclidean space. Suppose that $f(\boldsymbol{x})$ is analytic at the solution $\boldsymbol{r}^*$, $\nabla f(\boldsymbol{r}^*) = \mathbf{0}$ and $\nabla^2 f(\boldsymbol{r}^*)$ is nonsingular, we have

$$\lim_{i\to\infty} \inf \|\Phi(\boldsymbol{a}_i;\boldsymbol{r}^*)f(\boldsymbol{a}_i)\| > 0 \tag{3-33}$$

for any sequence $\{\boldsymbol{a}_i\} \subseteq \mathbb{X} \setminus \{\boldsymbol{r}^*\}$ converging to $\boldsymbol{r}^*$. Then, $\Phi(\boldsymbol{a}_i;\boldsymbol{r}^*)$ is called as a deflation function.

theorem 3.1

Suppose $f(\boldsymbol{x})$ is analytic at the solution $\boldsymbol{r}^*$, $\nabla f(\boldsymbol{r}^*) = \mathbf{0}$ and $\nabla^2 f(\boldsymbol{r}^*)$ is nonsingular. Let $\boldsymbol{a}_i$ be any sequence in $\mathbb{X}$ converging to $\boldsymbol{r}^*$, and let ω_i be any sequence in $\mathbb{X}$. Define $\|\cdot\|_*$ as a function of $\boldsymbol{a}_i - \boldsymbol{r}^*$. If

$$\|\boldsymbol{a}_i - \boldsymbol{r}^*\|_* \Phi(\boldsymbol{a}_i;\boldsymbol{r}^*)\omega_i \to 0 \quad \text{implies} \quad \omega_i \to 0 \quad \text{as} \;\; i \to \infty, \tag{3-34}$$

then Φ is a deflation function.

we first present several specific and distinct choices.

- When $\|\cdot\|_*$ is taken as a component of a vector, i.e., $\|\boldsymbol{a}-\boldsymbol{r}^*\|_* := (\boldsymbol{a}-\boldsymbol{r}^*)_j$, where $(\boldsymbol{a}-\boldsymbol{r}^*)_j$ denotes the j-th component of the vector $\boldsymbol{a}-\boldsymbol{r}^*$. Consequently, the corresponding deflation function can be chosen as

$$\Phi(\boldsymbol{a};\boldsymbol{r}^*) = \frac{1}{(\boldsymbol{a}-\boldsymbol{r}^*)_j} + \alpha, \tag{3-35}$$

  where $\alpha(>0)$ is a constant.

- When we set $\|\boldsymbol{a}-\boldsymbol{r}^*\|_* := \|\boldsymbol{a}-\boldsymbol{r}^*\|_a^p$ ($a = 1, 2, \infty$ and $p \geq 1$), the corresponding deflation function becomes

$$\Phi(\boldsymbol{a};\boldsymbol{r}^*) = \frac{1}{\|\boldsymbol{a}-\boldsymbol{r}^*\|_a^p} + \alpha. \tag{3-36}$$

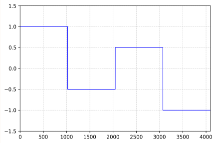
  It is worth pointing out that (3-36) agrees with the results shown in [3] (Farrell et al.).

Numerical experiments I

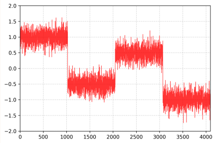
Example 1:

$$H^1-\textbf{denoised signal: } \min_{\boldsymbol{x}\in\mathbb{R}^n} f(\boldsymbol{x}) := \left\{\frac{1}{2}\|\boldsymbol{x}-\hat{\boldsymbol{x}}\|_2^2 + \lambda\|A\boldsymbol{x}\|_2^2\right\},$$

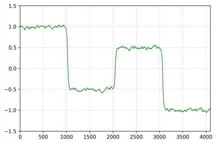
$$\text{TV-}\textbf{denoised signal: } \min_{\boldsymbol{x}\in\mathbb{R}^n} f(\boldsymbol{x}) := \left\{\frac{1}{2}\|\boldsymbol{x}-\hat{\boldsymbol{x}}\|_2^2 + \lambda\|A\boldsymbol{x}\|_1\right\}.$$



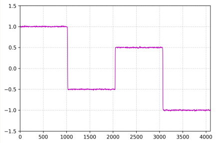
(d) Original signal



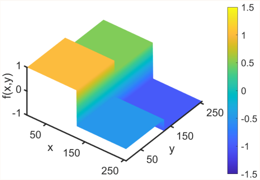
(e) Noisy signal



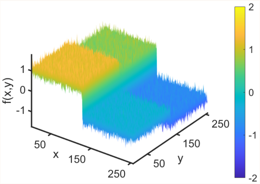
(f) H^1



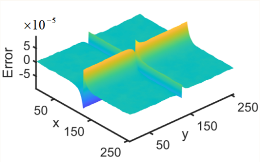
(g) TV



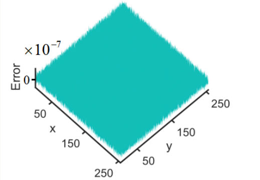
(h) Original signal



(i) Noisy signal



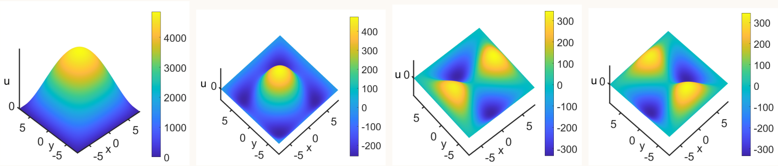
(j) H^1



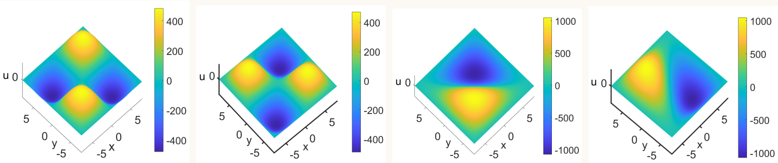
(k) TV

Example 2:

$$\begin{cases} -\triangle u + \lambda |u|^{2\alpha} u = 0, & \text{in } \Omega = (-8, 8) \times (-8, 8), \\ u = 0 & \text{on } \partial\Omega, \end{cases} \tag{4-37}$$



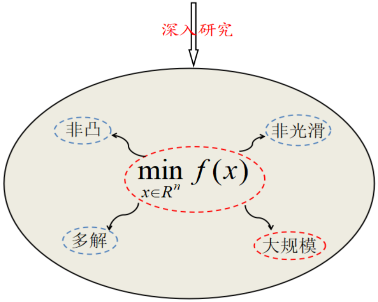
(l) I (m) II (n) III-1 (o) III-2



(p) IV-1 (q) IV-2 (r) V-1 (s) V-2

Figure 4-3 Multiple solutions of (4-37) with $\alpha = -0.3$ and $\lambda = -10$.

- **Algorithm 2** 在计算非光滑、非凸、多解等极小优化问题是非常有效的;
- 大规模？ 多解问题？



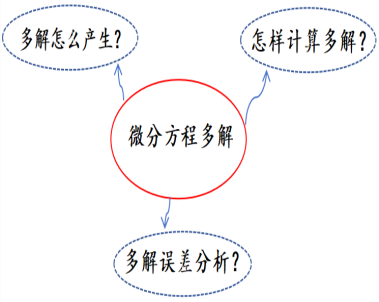


Figure 5-4 待研究的问题。

Thanks for your attention!